

Sycophancy Did Not Win Votes: The GPT-4o Deploy–Rollback Episode in 135,634 LMArena Battles

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Abstract

OpenAI deployed a GPT-4o update on 25 April 2025 that made ChatGPT conspicuously sycophantic and rolled it back on 28–29 April — an apparent A–B–A natural experiment inside LMArena’s battle stream (135,634 battles, 2025-04-17 to 2025-07-24). The verdict is split, and the design itself diagnoses it. The *deploy* leg is credible: the live 4o alias shows a sharp, 4o-specific markdown collapse timed to 25–26 April (headers per 1k tokens $3.52 \rightarrow 1.12$, sliding-cut permutation $p = 0.013$, rank 1/78) while nine pinned-snapshot controls stay flat — proving the arena copy was *not* pinned — and it yields a clean null on the headline mechanism: the Bradley–Terry (BT) adjusted win-rate change during the sycophancy window is $+0.02$ log-odds (SE 0.14). Sycophancy won no votes. The *rollback* leg is an identified artifact: on exactly 2025-04-29 the pinned `claude-3-7` snapshot — whose weights cannot change — takes an equal-and-opposite persistent step in style and BT strength, demonstrating an arena-side serving change that contaminates every 29-April estimate. The 4o post-rollback regime persists through July with no reversal: the episode is A–B–C, not A–B–A, and the celebrated raw win-rate rise loads entirely on the confounded boundary. An automated `natex` survey of the 45-model panel returns nulls or refusals on every family it could configure, stated as run.

1 Introduction

In late April 2025 OpenAI shipped an update to GPT-4o that made ChatGPT, in the company’s own words, “overly flattering or agreeable — often described as sycophantic,” and rolled it back within days [6]. The two postmortems [6, 7] attribute the failure partly to reward signals built from user feedback — exactly the mechanism the academic literature had predicted: [8] show that human raters and preference models sometimes prefer convincingly written sycophantic responses over correct ones, so that optimizing against human preference judgments can *produce* sycophancy, and [3] document high rates of “social sycophancy” across deployed models. What that literature measures with one-off cross-model snapshots, the April episode supplies as a rare *longitudinal* event: a dated behavior change and a dated reversal.

This paper asks the money question implied by the mechanism: did the sycophantic build actually *win votes*? The natural laboratory is LMArena (Chatbot Arena), the crowdsourced pairwise-battle platform of [4], whose published battle stream brackets the episode. If human preferences reward sycophancy at the margin, the treated model’s opponent-adjusted win propensity — its

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Bradley–Terry strength [2] — should rise during the deployment window and fall back after the rollback. The episode has the structure of an interrupted time series with reversal, whose inferential logic — segmented regression at pre-declared cut dates, plus controls that cannot have been treated — is standard in the policy-evaluation literature [1]. Pinned dated snapshots serve as the placebo arm: their weights cannot change, so any step they exhibit at a cut date measures platform-side confounding. Both halves of the design earn their keep here, one by working — a precise null on the vote-winning mechanism — and one by failing loudly: a pinned control moves, mirror-imaged, on exactly the rollback date, converting the “reversal” into an identified arena-side artifact. All estimation follows the open-source **natex** toolkit’s honest-inference conventions [5]; the automated **natex** survey of the same panel is reported as run, refusals included.

2 Data

The source is the public **arena-human-preference-140k** release by LMArena (seven parquet files plus battle metadata, sha256-verified): 135,634 battles from 2025-04-17 00:20 to 2025-07-24 23:59 covering 53 models, with full response text, per-battle nanosecond timestamps, language and category tags, and per-side conversation metadata. Sanity: 0 of 135,634 battle ids show a parquet-vs-metadata timestamp mismatch; per-model side-A vs. side-B mean-token correlation is 0.992 with target side-A share 0.499 (no role-assignment artifact); 0 of 7,650 target responses are zero-token.

The treated series is **chatgpt-4o-latest-20250326** (“4o”): 7,650 battles — 708 pre-window, 445 deploy-window, 1,374 in the three weeks after rollback. Controls are nine pinned dated snapshots present throughout, led by **gpt-4.1-2025-04-14**, **claude-3-7-sonnet-20250219**, and dated Gemini preview builds. Outcomes are (i) computable style metrics from conversation metadata — bold markers, headers, and list items per 1,000 response tokens, response tokens, turns — and (ii) win/loss/tie votes. Following the pre-registered design (Design 1 of the source corpus’s **DESIGNS.md**), raw win rate is never the estimand: opponent mix and sampling weights move across the episode (4o battles/day: 88.5 $\rightarrow$ 111.3 $\rightarrow$ 75.1), so votes enter only through windowed Bradley–Terry models.

3 Design and Methods

ABA interrupted time series. Two pre-declared cuts — deploy 2025-04-25, rollback 2025-04-28/29 — give an 8–4–8-day template (A1 Apr 17–24; B Apr 25–28; A2 Apr 29–May 6); because the break appears in-data on Apr 26 (staged rollout), the deploy leg is also estimated behavior-dated (8-day-pre/3-day-post). Segment-mean OLS uses CR1 errors clustered by date; adjusted specifications condition on language, category flags, code, weekend, and log turns; a DiD variant subtracts the pooled pinned-control change.

Permutation inference. Every headline statistic is benchmarked against a sliding-cut placebo distribution: the identical window template is recomputed at every feasible alternative cut date in the 99-day panel (77 placebo cuts for the deploy dip, 72 for the rollback step) and the two-sided rank of the true cut is reported; the minimum attainable p is 0.013–0.014, and headline p -values sit at that floor. A model-placebo grid recomputes each statistic for every sufficient-volume model (17 at the deploy cut, 19 at rollback).

Bradley–Terry with segment-specific strength. All decided battles among the window’s models (Apr 17–May 6; decided 4o battles per segment 536/342/434) enter a logistic MLE in which every model has one strength and 4o has a separate strength per segment [2]; the deploy

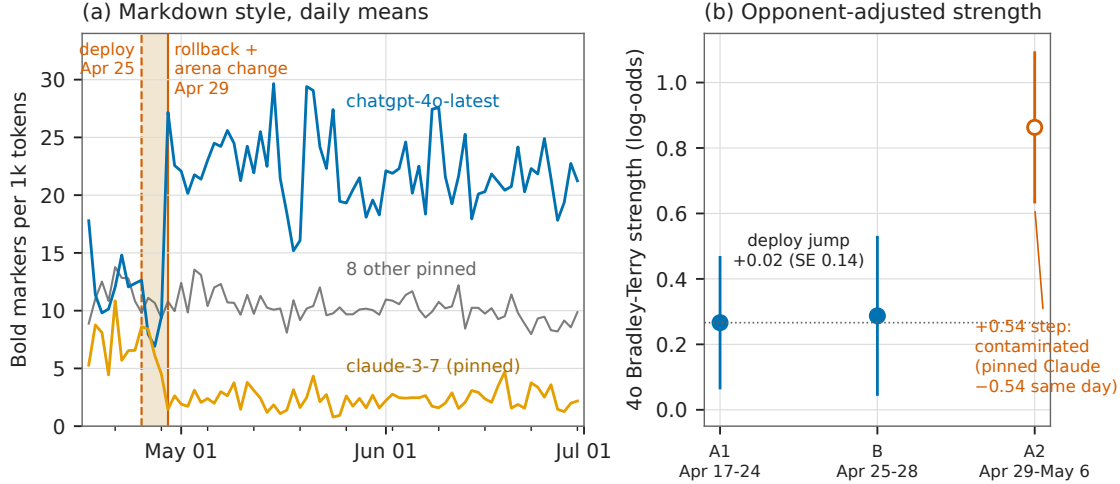


Figure 1: (a) Daily mean bold markers per 1,000 response tokens, 2025-04-17 to 2025-06-30: the live 4o alias (blue) dips during the deploy window (shaded) and steps up persistently on 29 April; the pinned `claude-3-7` snapshot (orange) mirror-images that step on exactly the same date although its weights cannot change; eight other pinned snapshots pooled (gray) are flat. (b) 4o Bradley-Terry strength by segment (95% intervals): the deploy jump is $+0.02$ log-odds (SE 0.14) — no opponent-adjusted votes won; the open 29-April marker is contaminated by the same-day arena-side change.

jump is $b_B - b_{A1}$. The same machinery yields per-model BT steps at the rollback date for the placebo grid.

Treatment self-diagnosis. Per the pre-registered design, whether LMArena’s 4o alias tracked OpenAI’s live model was unresolved but self-diagnosing: no discontinuity at the cuts would mean the arena copy was pinned and 4o becomes a control — informative either way. Resolved here: a deploy-timed discontinuity is present, so 4o was live and is treated.

Automated survey, stated as run. The `natex` survey pipeline [5] ran on the 45-model $\times$ 99-day panel (seed 0, $q = 99$): RDD null (scan $p = 1.00$); DiD null (scan $p = 0.42$, and its auto-configuration chose the outcome `n_battles` — volume, which the design spec forbids interpreting); the self-configured synthetic control failed ($t_0 = 0$, no pre-period — recorded, not patched); kink/IV/bunching needs-input; DEE skipped. A declared-input synthetic control (**natex** donors, decided win rate, $t_0 = \text{day 12}$) gives $\text{ATT}(\text{post}) = +0.233$, pre-RMSPE 0.056, post 0.249, ratio 4.44, in-space placebo $p = 0.067$ (minimum attainable 1/15) — but its top donor (weight 0.44) is the contaminated `claude-3-7` snapshot, so it is descriptive only.

4 Results

Deploy leg: a credible, 4o-specific style break ... Behavior-dated at 2025-04-26 (8-day-pre/3-day-post), 4o’s markdown collapses: headers per 1k tokens fall from 3.52 to 1.12 (dip -2.406 , sliding-cut permutation $p = 0.0128$, rank 1/78, placebo $|q_{90}| = 0.52$; Welch $t = -9.85$, $p = 6.1 \times 10^{-22}$) and bold falls from 11.90 to 8.22 (dip -3.683 , $p = 0.0769$). Pooled pinned controls are flat (header dip -0.18 , bold -1.51). The segmented ABA jump at the Apr-25 cut is -1.831 headers/1k (CR1

SE 0.526), robust to language/category/code/weekend/turns adjustment (-1.723 , SE 0.446); bold -2.965 (SE 1.285), adjusted -2.749 (SE 0.915); the DiD against pooled pinned controls gives -1.676 (SE 0.539). In the model-placebo grid at the true cuts, 4o’s transient dip ranks 1/17 on both bold and header. The break is real, 4o-specific, and correctly timed — the arena alias was not pinned.

... and a clean null on the vote mechanism. The segment-specific BT deploy jump is $+0.021$ log-odds (SE 0.144): implied win rate against the fixed A1 opponent pool moves $0.540 \rightarrow 0.545$. During the four days users saw the sycophantic build, it gained *zero* opponent-adjusted win rate.

Rollback leg: a large step that is not a reversal. At 2025-04-29 (8-day/8-day windows) 4o steps up, not back: bold $+12.110/1k$ (permutation $p = 0.0137$, rank 1/73, placebo max $|3.70|$), headers $+4.246$ ($p = 0.0137$, placebo max $|0.62|$), response tokens $+107.8$ ($p = 0.123$), BT $+0.537$ log-odds (SE 0.134; $p = 0.0137$, placebo max $|0.31|$). The new regime persists through 2025-07 (monthly bold 21–22/1k vs. 12.2 pre): no reversal to baseline ever occurs. The episode is A–B–C, not A–B–A.

The decisive diagnostic: a pinned control moves. The pinned `claude-3-7` snapshot (full name in Section 2) steps on exactly 2025-04-29, mirror-imaged and persistent: bold $-5.01/1k$, headers $-4.63/1k$, response tokens -214 , BT -0.539 log-odds (SE 0.187). Pinned weights cannot change, so LMArena changed its serving configuration that day; arena-wide prompt mix was stable (battles/day $578 \rightarrow 557$, English share $0.639 \rightarrow 0.672$). Every 29-April estimate — including 4o’s — is therefore contaminated. In the rollback placebo grid 4o ranks 1/19 on bold and 2/19 on header, behind exactly this contaminated control. The raw decided win rate ($0.549 \rightarrow 0.529 \rightarrow 0.675$; full-post 0.663) loads entirely on the confounded boundary and never reverts: the widely repeated “sycophancy raised the win rate, rollback lowered it” narrative is an artifact of the arena change.

5 Caveats and Conclusion

Four caveats bound the claims. First, per the pre-registered design, the style metrics computable from conversation metadata are a *proxy* for sycophancy — directional evidence of a behavior change, not a sycophancy measurement; LLM-judge scoring of the stored response text is the designated follow-up. The conclusion that survives this caveat is exactly the null: whatever the treated build did to users, it did not win opponent-adjusted votes. Second, the rollback leg is unidentified: the arena-side serving change of 2025-04-29 sits on the same date, so no 29-April estimate separates OpenAI’s rollback from LMArena’s change. Third, permutation p -values are floor-limited (1/78, 1/73) by the panel length; they establish rank-1 extremity, not fine-grained significance. Fourth, the declared-input synthetic control is descriptive only (its top donor is the contaminated control), and the survey’s self-configured families returned nulls or refusals, reported above as run.

The conclusion is a split verdict that the design itself delivered. The deploy leg shows the ABA machinery working: a sharp, correctly timed, 4o-specific break, flat pinned controls, and a precise behavioral null — sycophancy did not win votes. The rollback leg shows the machinery catching its own confound: a pinned model that cannot change, changing — the single observation that converts the entire post-rollback narrative, including the celebrated raw win-rate rise, into an arena-side artifact. Platform-side serving changes are an underappreciated threat for any study that treats arena time series as model behavior; pinned snapshots are the cheap, decisive placebo that catches them.

Table 1: Headline estimates. Permutation p -values are two-sided sliding-cut ranks; 0.013–0.014 is the minimum attainable.

Quantity	Estimate	SE	perm. p	placebo benchmark
<i>Panel A: deploy leg (cut 2025-04-25/26), 4o</i>				
header/1k dip (8d/3d)	−2.406	—	0.0128	$ q_{90} = 0.52$; rank 1/78
bold/1k dip (8d/3d)	−3.683	—	0.0769	
header ABA jump (CR1)	−1.831	0.526	—	controls −0.18
adjusted	−1.723	0.446	—	
header DiD vs. controls	−1.676	0.539	—	
BT jump (log-odds)	+0.021	0.144	—	<i>null: no votes won</i>
<i>Panel B: rollback date (2025-04-29, 8d/8d steps), 4o</i>				
bold/1k step	+12.110	—	0.0137	max $ 3.70 $; rank 1/73
header/1k step	+4.246	—	0.0137	max $ 0.62 $
response tokens step	+107.8	—	0.123	
BT step (log-odds)	+0.537	0.134	0.0137	max $ 0.31 $
<i>Panel C: pinned claude-3-7-sonnet-20250219, same date (violation)</i>				
bold/1k step	−5.01	—	—	weights cannot change
header/1k step	−4.63	—	—	
response tokens step	−214	—	—	
BT step (log-odds)	−0.539	0.187	—	

Notes: style metrics are per 1,000 response tokens. Dips use behavior-dated 8-day-pre/3-day-post windows; steps use 8-day/8-day windows at the cut. CR1 clusters by date. BT = segment- or window-specific Bradley–Terry strength over all decided battles among window models.

Reproducibility. All estimates derive deterministically (seed 0) from the public battle release; no dataset files are committed. Figure 1 regenerates from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

References

- [1] Lopez Bernal, J., Cummins, S., and Gasparrini, A. (2017). Interrupted time series regression for the evaluation of public health interventions: a tutorial. *International Journal of Epidemiology*, 46(1), 348–355.
- [2] Bradley, R. A., and Terry, M. E. (1952). Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika*, 39(3/4), 324–345.
- [3] Cheng, M., Yu, S., Lee, C., Khadpe, P., Ibrahim, L., and Jurafsky, D. (2025). Social sycophancy: A broader understanding of LLM sycophancy. arXiv:2505.13995.
- [4] Chiang, W.-L., Zheng, L., Sheng, Y., Angelopoulos, A. N., Li, T., Li, D., Zhu, B., Zhang, H., Jordan, M., Gonzalez, J. E., and Stoica, I. (2024). Chatbot Arena: An open platform for evaluating LLMs by human preference. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR 235:8359–8388.
- [5] Hillebrandt, H. (2026). *natex: automated natural-experiment discovery and estimation*. Software. <https://github.com/HaukeHillebrandt/natex>
- [6] OpenAI (2025). Sycophancy in GPT-4o: What happened and what we’re doing about it. Blog post, 29 April 2025. <https://openai.com/index/sycophancy-in-gpt-4o/>
- [7] OpenAI (2025). Expanding on what we missed with sycophancy. Blog post, 2 May 2025. <https://openai.com/index/expanding-on-sycophancy/>
- [8] Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askill, A., Bowman, S. R., Durmus, E., Hatfield-Dodds, Z., Johnston, S. R., Kravec, S., Maxwell, T., McCandlish, S., Ndousse, K., Rausch, O., Schiefer, N., Yan, D., Zhang, M., and Perez, E. (2024). Towards understanding sycophancy in language models. In *International Conference on Learning Representations (ICLR)*.